Hidden Markov Model Analysis of Oncocell-Binding Protein Mechanisms in the Maraba Virus

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Introduction

Oncolytic viruses have emerged as a promising class of cancer therapeutics due to their unique ability to selectively infect, replicate within, and lyse malignant cells while sparing healthy tissues. Among the diverse viral families explored for oncolytic potential, *Vesiculoviruses* (negative-sense, single-stranded RNA viruses within the order *Mononegavirales*) have gained increasing attention for their potent cytolytic activity and immunostimulatory properties (Lawler et al., 2017). One of the most notable members of this group is the maraba virus, a naturally occurring rhabdovirus originally isolated from Brazilian sandflies. Genetically engineered derivatives of the maraba virus such as MG1 have demonstrated strong tumor selectivity and the capacity to enhance antigen presentation and anti-tumor immune responses, making maraba a leading platform for oncolytic virotherapy research (Brun et al., 2010).

The genomic architecture of *Vesiculoviruses* consists of a linear ~11 kb genome that encodes five major proteins (N, P, M, G, and L), arranged in a highly conserved order. Of these, the glycoprotein (G) plays a central role in determining host-cell entry, tissue tropism, and immunogenicity. Differences in G-protein sequence and structure have been implicated in the enhanced cancer-specific recognition observed in oncolytic strains such as *Vesiculovirus* maraba compared to their non-oncolytic relatives (Pol et al., 2014). Previous studies have identified specific point mutations, including Q242R, that modulate tumor-cell binding and may underlie differences in therapeutic efficacy between closely related viral species (Brun et al., 2010). Understanding how variation in the G protein relates to oncolytic potential is therefore critical for determining the molecular features that distinguish clinically promising viruses.

Comparative genomics offers multiple complementary approaches for investigating these differences. Synteny analyses reveal broad patterns of genome conservation and rearrangement, enabling identification of genomic regions that differ substantially between oncolytic and non-oncolytic species. Multiple sequence alignments (MSAs) of viral glycoproteins provide residue-level insight into conserved motifs, structural domains, and mutational hotspots that may contribute to oncolytic activity. Beyond traditional alignment-based methods, statistical models such as Hidden Markov Models (HMMs) offer a powerful means of capturing position-specific variability and inferring consensus patterns that distinguish groups of related sequences. HMMs have been widely applied in gene prediction, protein family classification, and motif detection due to their ability to model insertions, deletions, and residue probabilities in a biologically meaningful way (Yoon, 2009).

However, despite extensive use of HMMs in molecular sequence analysis, their application to distinguishing oncolytic from non-oncolytic *Vesiculovirus* species based on glycoprotein sequence patterns remains largely unexplored. Existing tools such as HMMER3 provide robust frameworks for profile HMM construction, but their complex architectures can obscure interpretable features when applied to small viral datasets. Developing a customized, biologically interpretable HMM framework therefore provides an opportunity to examine whether consistent sequence signatures differentiate the glycoproteins of oncolytic strains from those of closely related, non-oncolytic viruses. Integrating HMM-based sequence modeling with syntenic, phylogenetic, and structural insights can help reveal the evolutionary and functional mechanisms driving tumor specificity in *Vesiculoviruses*.

Given the promise of the maraba virus as an oncolytic agent and the clear gaps in our understanding of the molecular determinants of its tumor selectivity, a combined statistical and comparative genomics approach may illuminate how unique sequence features in the G protein contribute to the virus's therapeutic capacity. This study investigates these questions through the development and testing of custom HMMs trained on glycoprotein MSAs, combined with synteny comparisons and phylogenetic analyses across multiple *Vesiculovirus* species.

Objectives

The objective of this study is to investigate the molecular features that distinguish oncolytic *Vesiculovirus* species from their non-oncolytic relatives, with a specific focus on glycoprotein sequence variation. This project is guided by the following goals:

1. To characterize conserved and variable sequence features of oncolytic and non-oncolytic *Vesiculovirus* glycoproteins by generating and analyzing multiple sequence alignments (MSAs) to identify motifs, structural domains, and residue patterns that may contribute to functional divergence.
2. To construct both simplified (lightweight) and biologically accurate HMMs trained on glycoprotein MSAs from oncolytic and non-oncolytic viral groups, enabling examination of the statistical signatures that differentiate these two categories.
3. To evaluate the performance of the custom HMMs by benchmarking against the established HMMER3/pyHMMER framework, assessing the degree of agreement in scoring profiles and validating the reliability of the custom models.
4. To test classification accuracy through a leave-one-out approach, determining whether held-out glycoprotein sequences can be correctly assigned to their respective oncolytic or non-oncolytic groups based solely on HMM-derived log-likelihood scores.
5. To compare genome-wide synteny between oncolytic and non-oncolytic *Vesiculovirus* species, with particular emphasis on divergence in the M and G gene regions that may correlate with differences in host recognition and tumor selectivity.

6. To integrate HMM-based sequence modeling with phylogenetic and synteny analyses to develop a cohesive understanding of how glycoprotein variation and genomic structure relate to oncolytic potential across the *Vesiculovirus* genus.

Methods

Pairwise Synteny Alignments

A literature review was conducted in order to determine likely gene candidates for oncolytic activity. The team of Brun et al. (2010) used shotgun methods to introduce a point mutation in the maraba genome and later check their oncolytic activity in vitro to determine the virus strain with the highest oncolytic activity. Afterwards a titer analysis was performed in order to ensure that the virus selected for its oncolytic activity was able to infect and reproduce efficiently even with these introduced point mutations. Later sequencing of the selected virus found that the M and G (Q242R) genes in the virus, dubbed maraba MG1, were likely responsible for the increased activity (Brun et al., 2010). The M gene role in increased activity was likely due to downstream genetic effects that were too complicated to measure for the sake of this research (Brun et al., 2010). However, the G gene directly codes for the G protein receptor on the surface of the virus. Cancer cells are known to overexpress these receptors which is likely one method the maraba virus uses to located and target cancer cells (Elnosary et al., 2024). To be sure of the gene selection Artemis ACT was used to prepare a synteny alignments between maraba virus (NCBI Taxonomy ID: 1972574) and a slightly less oncolytic related species, *Vesiculovirus carajas* (NCBI Taxonomy ID 1972571) (Brun et al., 2010). Areas with less synteny were deemed likely candidates for increased oncolytic activity (Appendix 1).

A second synteny alignment was performed that compared the maraba virus with a completely non-oncolytic species, *Vesiculovirus radi* (NCBI Taxonomy ID: 1972566), in order to confirm the findings in the first synteny alignment (Appendix 2) (Elliott et al., 2025). In both synteny alignments, the areas with the least synteny are the M and G genes. The G gene was selected as the most direct influence on oncolytic activity.

Multiple Sequence Alignment Construction

The G gene sequence of the maraba virus was then extracted as an amino acid FASTA file and compared to the NCBI database using BLAST to find species possessing G proteins with varying degrees of similarity for the multiple sequence alignment (MSA). From these related species, two MSAs were generated using Seaview; one using a selection of oncolytic virus species and another using non-oncolytic viruses (Appendix 3).

Once the HMMs were prepared, these MSAs would be used to train them. These MSAs were additionally used to construct a potential phylogenetic tree based on the G protein amino acid sequences (Appendix 4). This taxonomic tree was used both to confirm the results of the pairwise synteny alignments by comparing the ancestral relatedness of the maraba virus, *Vesiculovirus carajas*, and *Vesiculovirus radi*; as well as to select the viral species to be removed from the training set when performing the leave-one-out tests on the HMMs.

Building the HMM Framework

We chose to implement a few different features when designing our HMM framework to capture how genes are created and mutated. We designed two different frameworks, a lightweight model, and a more biologically accurate model. Both models contain Match States and a Global Insertion State, the minimum requirements for learning the patterns in a family of genes. The biologically accurate model also contains a global deletion state, variable length insertions/deletions and a base probability for absent proteins. Below we discuss what each feature is and how it was implemented, followed how we validated our models.

Match States

Match states are the basic building blocks of HMMs. An HMM can be represented by a sequence of match states, with each match state representing the probabilities of each protein at each position. In order to create each match state, we looked at the position specific protein probabilities for each protein and set these to emission probabilities of each match state. When choosing which positions were used as match states, we looked at the training MSA and chose positions that had >50% of the sequences with a protein, the remaining positions were then considered to have undergone an insertion. This allowed us to calculate the transition probabilities from a match state to the next match or to an insertion/deletion. We calculated the number of insertions that occurred in our MSA and divided it by the total number places where an insertion could have occurred, this gives us our base insertion probability that we used for our transition probabilities.

Global Insertion State

We chose to use a global insertion state rather than position specific insertions because we know that insertions can occur at any position with an equal probability. Furthermore, it helps make the model slightly more efficient. When calculating the emission probabilities for our insertion state, we simply looked at whenever an insertion had occurred and looked at the proportion that each protein was inserted. For the transition probabilities we used the same base insertion (and deletion, when applicable) probabilities of the match states.

Global Deletion State

Similarly as above, we opted for a global deletion state. Deletions were detected based on the gaps in positions that were labelled as a match state. Deletion states are silent and thus have no emission probabilities. The transition probabilities out of the global deletion state are the same as the global insertion states.

Variable Length Insertion/Deletion (InDel) Sizes

Traditionally, InDels are modelled one at a time, with an increased probability to occur again to represent InDels that are larger than one protein long. We opted to represent this more biologically, we know that InDels can be any length, and so we allowed for this, in order not to get too out of whack with a random full deletion, or insertion of length 500, we chose to sample InDel length based on the proportion of each size we observed. In other words, if 50% of insertions were five proteins longs, the model would have a 50% chance, when inserting, to insert five proteins. One downside to adding this is the increased computational cost when scoring observations against the HMM. The Viterbi Algorithm is normally ~ $O(n^2)$, but because when we calculate the probability each path we have to each possible InDel size this adds an extra degree of time complexity and so the scoring process becomes ~ $O(n^3)$. This causes the algorithm to potentially become quite slow for longer sequences.

Protein Base Rates

As mentioned above, when calculating our emission probabilities, we simply used the observed proportion for each protein. While this is effective, it can lead to a rigid and overtrained model. To add flexibility and account for point-mutations, we added a tiny non-zero probability for proteins that were not observed in a position. This made our model slightly more flexible and should help it detect similar proteins that were not necessarily in the training data.

Scoring Algorithm

One of the most beneficial uses of an HMM is its ability to score an observed sequence against its learned statistical patterns, this lets the user determine how likely a sequence is to have come from the same family that the HMM was trained on. We implemented the Viterbi Algorithm for this. It is an algorithm that utilizes dynamic programming to iteratively calculate the probability of the sequence, one position at a time. It starts at the first position and determines what the most likely state is (Match, Insertion, or Deletion) based on what protein was observed. Then it looks at the protein in the next state and determines what the probability of this protein occurring is based on the previous most likely state transitioning to a new state and emitting this protein. It does this for each position until a final sequence probability is calculated which is the final score.

Model Validation

To ensure that our custom HMM frameworks were made properly and can be used for our glycoprotein analysis, we measure its performance against that of the HMMER model maintained by the Eddy Lab at Harvard (Eddy, 2011) implemented in python using the package pyHMMER (Larralde et al., 2022). We trained 3 HMMs on a sample MSA of 6 sequences that were ~40 proteins long. The three HMMs are as follows; Our lightweight framework, our biologically accurate framework and finally the pyHMMER framework. We then generated 50 sequences from each of our own models and scored them against the model they came from and the pyHMMER model. We found, for both of our frameworks, that all the sampled sequences were detected by the pyHMMER model.

Furthermore, we compared each respective model's scores to those from the pyHMMER model. We can see that for the lightweight model that there is a positive correlation, however there are some outliers (Appendix 5). These outliers likely represent sequences that are seen as quite unlikely by the lightweight model likely due to a lack of some important feature, like a deletion. Most notably our biologically accurate model had a very strong positive correlation with the scores from the pyHMMER model (Appendix 6). This means sequences that our model found likely to be from its HMM, the pyHMMER model also found to be likely from its model and this was also the case for less likely sequences. While this does not guarantee that our model is the same as the pyHMMER model, it indicates that our model is working well and able to properly capture the statistical patterns in an MSA.

Results

Consensus Sequence Profiles

Profile hidden Markov models (HMMs) were trained on the non-oncolytic and oncolytic vesiculovirus glycoprotein MSAs, and each model was used to generate a consensus sequence representing the most probable amino acid at each position. The non-oncolytic consensus sequence (Appendix 7A) began with MMILLAAICLIIIVSCSACALPIVFPDQR and had a log-likelihood of -348.6, indicating strong agreement with the emission patterns learned from the non-oncolytic training alignment. In contrast, the oncolytic consensus (Appendix 7B) began with QMLFALLLLCLLALGAHCKFTIVFPQSQK and produced a log-likelihood of -284.7, showing similarly strong support under the oncolytic HMM. Across the first 90 positions of each consensus (Appendix 7A & 7B), the two groups displayed clear sequence differences. These differences reflect distinct residue preferences encoded by each model and suggest potential functional divergence between oncolytic and non-oncolytic glycoproteins.

Leave-One-Out HMM Classification

To evaluate whether the HMMs could correctly classify sequences that we excluded from training, one oncolytic virus (*Vesiculovirus carajas*) and one non-oncolytic virus (*Vesiculovirus perinet*) were withheld. When tested against both models (Appendix. 8), *Vesiculovirus carajas* scored -7317.6 under the non-oncolytic model and -749.4 under the oncolytic model resulting in correct classification as oncolytic, while *Vesiculovirus perinet* scored -6373.4 under the non-oncolytic model and -7853.5 under the oncolytic model resulting in correct classification as non-oncolytic. These results show that the HMMs captured distinct sequence features even when individual sequences were removed from the training MSAs.

Full Dataset HMM Scoring

All glycoprotein sequences were then scored against both trained HMMs to compare likelihood distributions between the two virus groups (Appendix. 9). Oncolytic glycoproteins consistently produced higher log-likelihoods under the oncolytic HMM and lower scores under the non-oncolytic HMM. The inverse pattern was seen for all non-oncolytic sequences. This indicates that both HMMs captured non-overlapping sequence patterns that distinguish the two classes of vesiculoviruses.

Discussion

This study set out to determine whether glycoprotein sequences from oncolytic and non-oncolytic vesiculoviruses contain consistent molecular signatures that help explain their differences in tumor selectivity. Using a combination of synteny analysis, multiple sequence alignments, and custom hidden Markov model (HMM) frameworks, we examined both large-scale genomic patterns and residue-level variation. The synteny results highlighted divergence in the M and G gene regions between oncolytic and non-oncolytic vesiculoviruses, supporting the decision to focus on the G protein as a likely contributor to functional differences. HMM-derived consensus sequences further revealed distinct amino acid preferences, particularly within the early N-terminal region. Importantly, both HMM frameworks consistently separated the two viral groups, and the leave-one-out validation accurately classified withheld sequences. Together, these findings show that measurable sequence features distinguish oncolytic vesiculoviruses from non-oncolytic vesiculoviruses and can be captured using computational approaches.

The sequence differences observed between oncolytic and non-oncolytic vesiculoviruses, particularly within the N-terminal portion of the glycoprotein, suggest meaningful variation in regions that influence viral entry and host-cell interactions. The early N-terminal domain of rhabdovirus glycoproteins includes residues involved in signal peptide processing and membrane trafficking; shifts in amino acid preferences at these positions may alter how efficiently the G protein is folded, presented, or recognized by host receptors (Albertini et al., 2012). The fact that oncolytic sequences displayed consistent divergence from non-oncolytic sequences in this region supports the idea that tumor selectivity may arise from subtle changes affecting receptor affinity or entry efficiency. These patterns also align with earlier observations that specific point mutations can modulate oncolytic activity, further indicating that the glycoprotein gene represents a key locus of functional adaptation among vesiculoviruses (Brun et al., 2010).

The next step of this research would be to directly compare the oncolytic and non-oncolytic HMM profiles. The best approach for this would be to compare the protein probability distributions for each match state and examine the differences. Since our sequences have more than 500 proteins, we should highlight which positions varied the most between oncolytic and non-oncolytic glycoproteins.

Using this information, we can then investigate the functional differences between oncolytic and non-oncolytic glycoproteins. A structural analysis of the most variable regions would help us understand how functional differences emerge. From a successful functional analysis, we would be able draw conclusions about what makes the glycoproteins oncolytic.

Additionally, a limitation of our work is that the HMMs we trained could have been made more specific and reliable by using improved MSAs to train them. The MSAs used to train the HMMs used for this project used only a handful of species each. Increasing the number of species used in the MSAs would provide the HMMs with more training data, thereby improving their performance.

References

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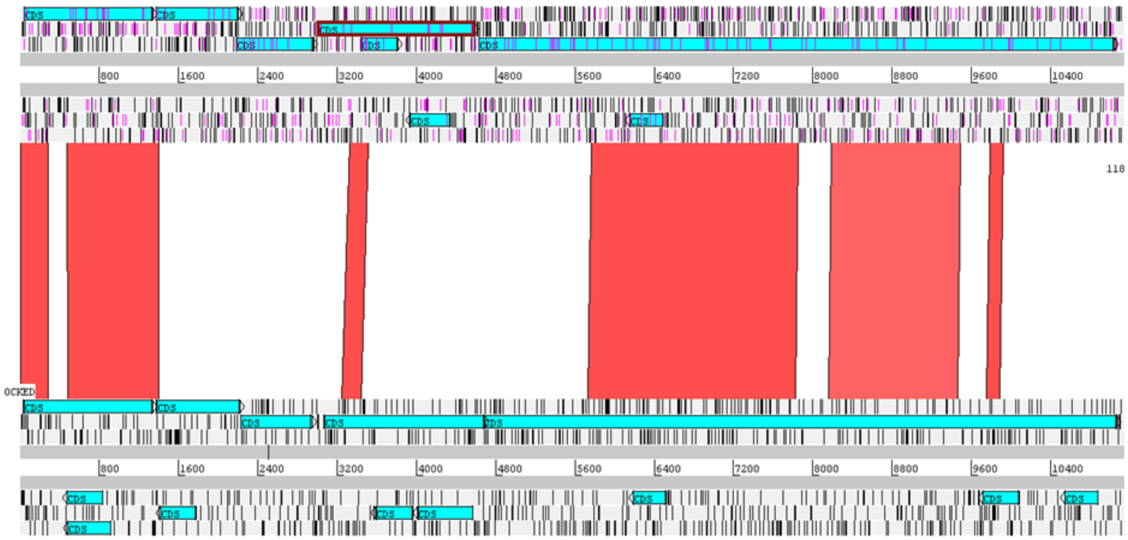
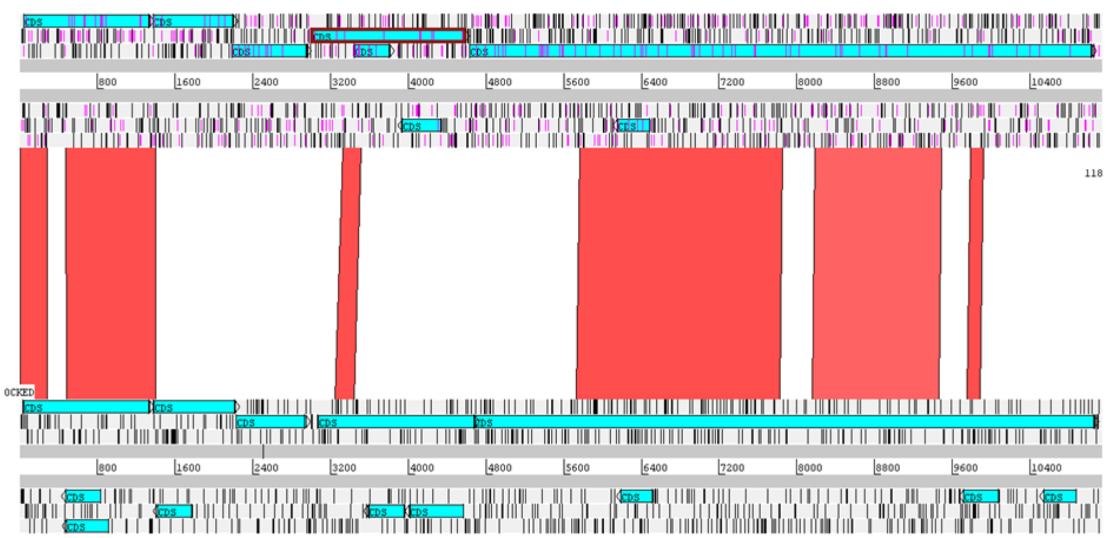
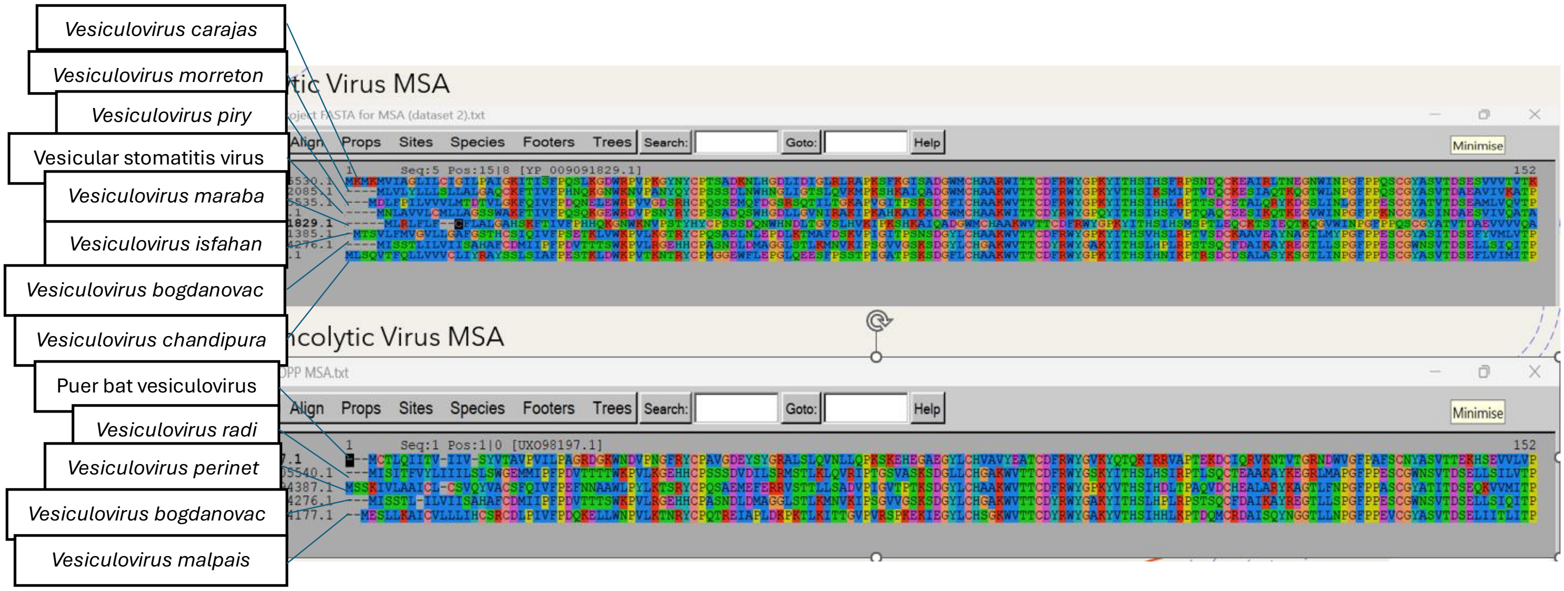
Appendix



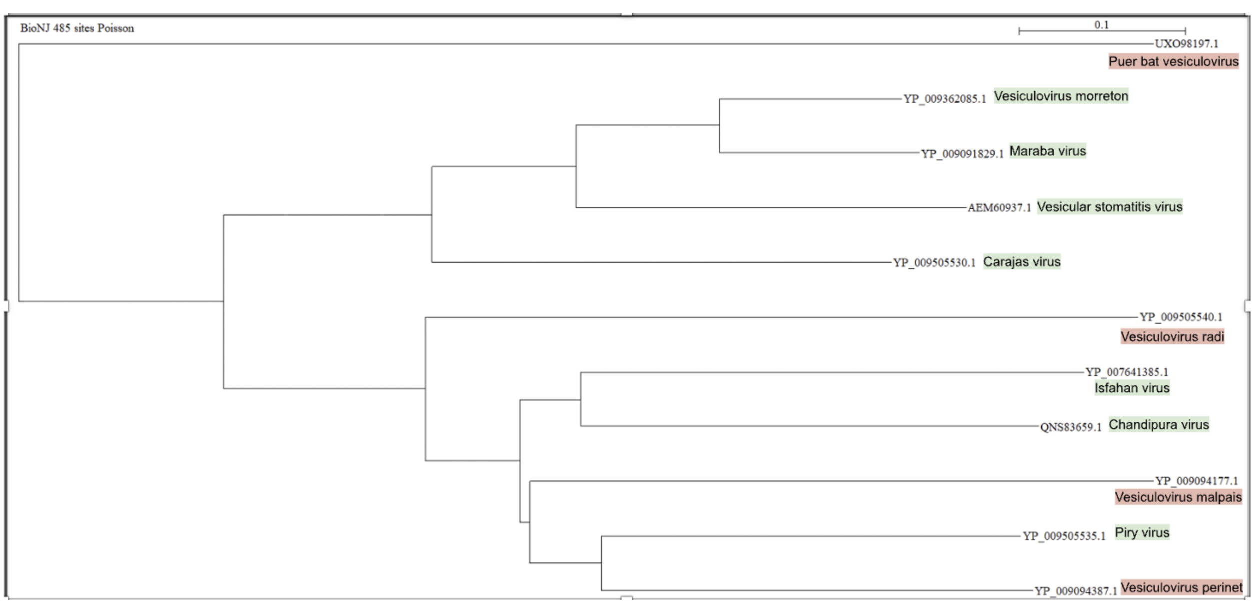
Appendix 1: Synteny alignment of *Vesiculovirus maraba* genome (top) and *Vesiculovirus carajas* genome (bottom). The vertical red bars indicate areas of high synteny between the two genomes. The open reading frames of the genes (N, P, M, G, and L) are indicated by horizontal cyan bars.



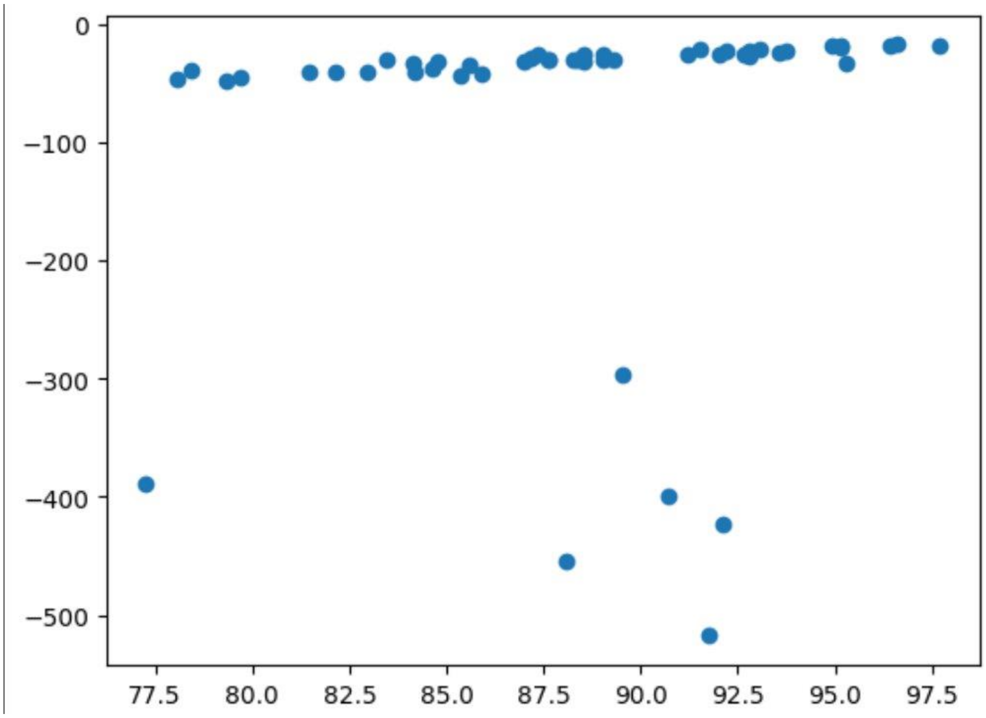
Appendix 2: Synteny alignment of *Vesiculovirus maraba* genome (top) and *Vesiculovirus radi* genome (bottom). The vertical red bars indicate areas of high synteny between the two genomes. The open reading frames of the genes (N, P, M, G, and L) are indicated by horizontal cyan bars.



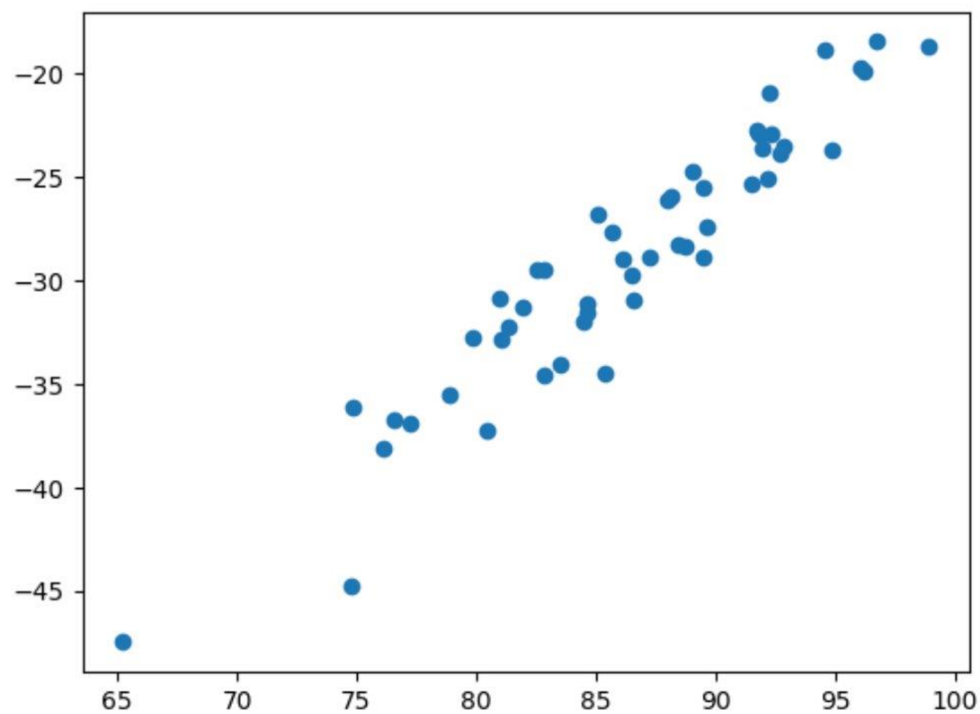
Appendix 3: Oncolytic (upper) and non-oncolytic (lower) virus G protein MSAs generated using Seaview. The organisms used and their positions in each MSA are listed to the left. *Vesiculovirus bogdanovac* was later removed from both MSAs due to a lack of concrete research into the oncolytic activity of the species.



Appendix 4: Phylogenetic tree constructed from both oncolytic and non-oncolytic virus MSAs. Oncolytic virus species are highlighted in green while non-oncolytic viruses are highlighted in red.



Appendix 5: Lightweight model scores (y-axis) vs pyHMMER scores (x-axis).



Appendix 6: Biologically model scores (y-axis) vs pyHMMER scores (x-axis).

```
...
    ==============================
       Non-oncolytic glycoproteins
    ==============================
                                                        0  1  2  3  4  5  6  7  8   \
    UX098197.1 glycoprotein [Puer bat vesiculovirus]    -  -  -  M  C  T  L  Q  I
    YP_009505540.1 glycoprotein [Vesiculovirus radi]    -  -  -  M  I  S  I  T  F
    YP_009094387.1 glycoprotein [Vesiculovirus peri...  M  S  S  K  I  V  L  A  A
    YP_009094177.1 glycoprotein [Vesiculovirus malp...  -  -  M  E  S  L  L  K  A

                                                        9   ... 34 35 36 37 38 39  \
    UX098197.1 glycoprotein [Puer bat vesiculovirus]    I  ...  W  N  D  V  P  N
    YP_009505540.1 glycoprotein [Vesiculovirus radi]    V  ...  W  K  P  V  L  K
    YP_009094387.1 glycoprotein [Vesiculovirus peri...  I  ...  W  L  P  Y  L  K
    YP_009094177.1 glycoprotein [Vesiculovirus malp...  I  ...  W  N  P  V  L  K

                                                        40 41 42 43
    UX098197.1 glycoprotein [Puer bat vesiculovirus]    G  F  R  Y
    YP_009505540.1 glycoprotein [Vesiculovirus radi]    G  E  H  H
    YP_009094387.1 glycoprotein [Vesiculovirus peri...  T  S  R  Y
    YP_009094177.1 glycoprotein [Vesiculovirus malp...  T  N  R  Y

    [4 rows x 44 columns]

    Cons (first 90): MMILLAAICLIIIVSCSACALPIVFPDQRNAAWNPVLKGNRYCPQSADEAILRRASTLKLNARVPIGSEAEKSDGYLCHAAKWVTTCDFR
    Samp (first 90): -EIVLTFVTLIIILHYSRCSMPIPLPDQRTALWKDYLNTNRYCPQVSEVDPEGKMKTLKLTARVPIGSEAEKAEGYLCHAAKYVATCDFR

    Viterbi on sample:
      logP: -5770.841329545448
      len(obs): 529  len(path): 529
      path head: MMMIMIMMIMMMMMIIIMMIIMIIIIMIMMIMMMIMIMIIIMMIIIMIIMMMMIIMMMIM

    Viterbi on consensus:
      logP: -348.57949593386144
      len(obs): 537  len(path): 537
      path head: MMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMM
```

(A)

```
    ==============================
       Oncolytic glycoproteins
    ==============================
                                                        0  1  2  3  4  5  6  7  8   \
    YP_009505530.1 glycoprotein [Carajas virus]         M  K  M  K  M  V  I  A  G
    YP_009362085.1 glycoprotein [Vesiculovirus morr...  -  -  -  -  M  L  V  L  Y
    YP_009505535.1 glycoprotein [Piry virus]            -  -  -  M  D  L  F  P  I
    AEM60937.1 glycoprotein [Vesicular stomatitis v...  -  -  -  -  M  N  L  A  V

                                                        9   ... 34 35 36 37 38 39  \
    YP_009505530.1 glycoprotein [Carajas virus]         L  ...  W  R  P  V  P  K
    YP_009362085.1 glycoprotein [Vesiculovirus morr...  L  ...  W  K  N  V  P  A
    YP_009505535.1 glycoprotein [Piry virus]            L  ...  W  R  P  V  V  G
    AEM60937.1 glycoprotein [Vesicular stomatitis v...  V  ...  W  R  D  V  P  S

                                                        40 41 42 43
    YP_009505530.1 glycoprotein [Carajas virus]         G  Y  N  Y
    YP_009362085.1 glycoprotein [Vesiculovirus morr...  N  Y  Q  Y
    YP_009505535.1 glycoprotein [Piry virus]            D  S  R  H
    AEM60937.1 glycoprotein [Vesicular stomatitis v...  N  Y  R  Y

    [4 rows x 44 columns]

    Cons (first 90): QMLFALLLLCLLALGAHCKFTIVFPQSQKGNWKPVPKNYRYCPSSADQNLHGDLIGTSLRSKAPKGHKAIKADGWMCHAAKWVTTCDFRW
    Samp (first 90): --LLRGGVVLCGILSSICKFTISFPEQTKGEWKPVVKNYQYCPSSSEMNWEPGLITVGFHGRIPIGITPSNSDGWMCHAAKWITTCDFRW

    Viterbi on sample:
      logP: -5612.68603801198
      len(obs): 532  len(path): 532
      path head: MIIMIMMMMMMMIMMIIIMIMMMMIIMIIMMIIIIMMMMMMMMMMMMMMMMIIIMIIMI

    Viterbi on consensus:
      logP: -284.6842797427322
      len(obs): 520  len(path): 520
      path head: MMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMM
```

(B)

Appendix 7: (A) Non-oncolytic consensus sequence with HMM alignment output and log-likelihood score. (B) Oncolytic consensus sequence with corresponding HMM alignment output and log-likelihood score.

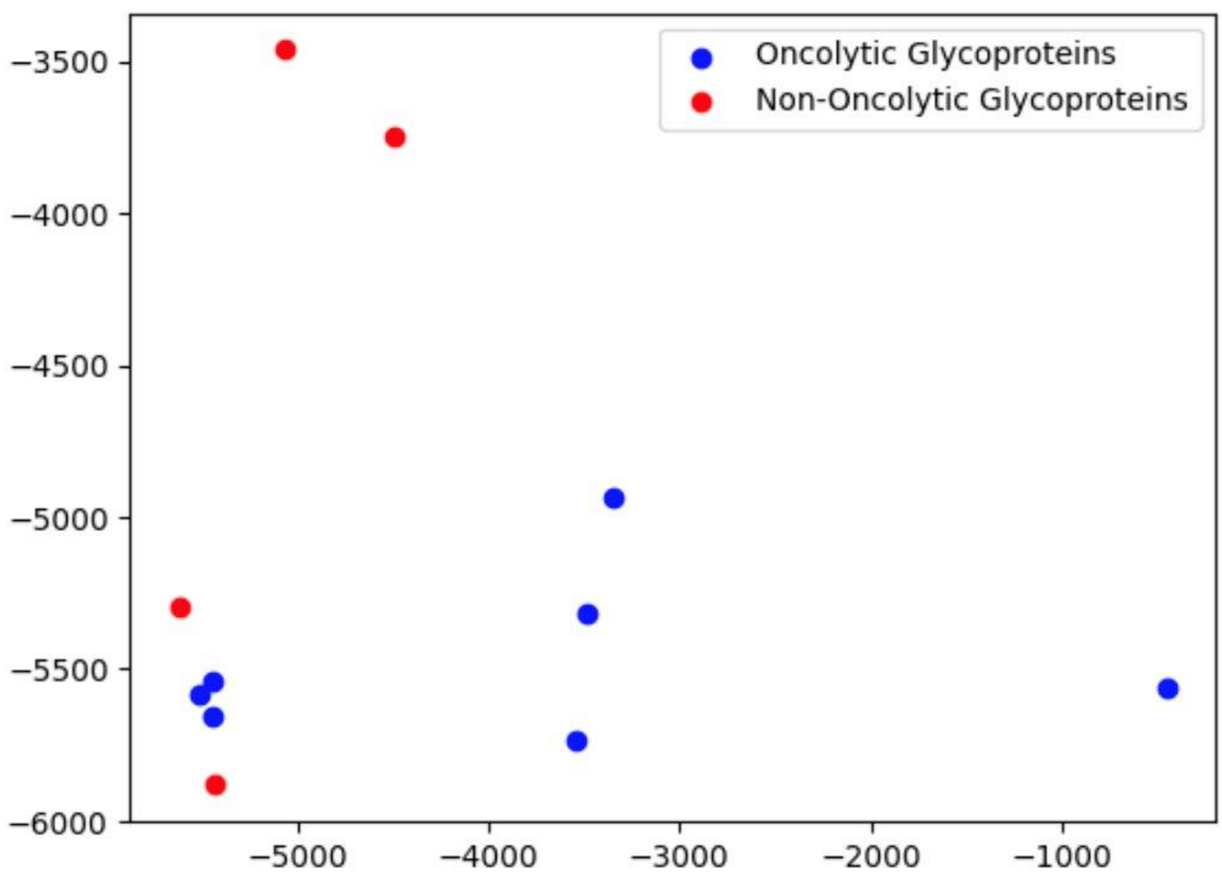
```
... Non-oncolytic HMM match states: 521
    Oncolytic   HMM match states: 512

    Non-oncolytic consensus (60 aa): MCSLQIAICIIILIHCARCDIIPAGRDQKENDKPNGLRENRACGDEREDARARKLQLNLQ
    Oncolytic     consensus (60 aa): MNRAMVICGLALGAGIAKAIGKIHQSQGNELRDDPRNHRCCPSSADNSANDNLGGDLIRA

    Carajas (oncolytic test):
      length: 523
      logP under NON-oncolytic HMM: -7317.5681344883515
      logP under ONCOLYTIC   HMM: -749.4225715104845

    Vesiculovirus perinet (non-oncolytic test):
      length: 525
      logP under NON-oncolytic HMM: -6373.416584244583
      logP under ONCOLYTIC   HMM: -7853.49653938646
```

Appendix 8: Log-likelihood comparison for withheld oncolytic and non-oncolytic sequences under both models (derived from leave-one-out scoring outputs).



Appendix 9: Scatterplot of oncolytic vs. non-oncolytic HMM scores for all sequences. Blue points represent oncolytic glycoproteins; red points represent non-oncolytic glycoproteins.